



Lecture 23.07: Starting with Linear Regression

 Python 3.10+

 Scikit-Learn 1.4+

Status Completed

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1. What is Linear Regression?

Linear Regression is a fundamental supervised learning algorithm that models the relationship between a continuous scalar target variable y and one or more explanatory feature variables $\mathbf{x}$ by fitting a linear equation to observed data.

1. Simple Linear Regression (Single Feature x):

$$\hat{y} = wx + b \quad \text{or} \quad \hat{y} = \beta_1 x + \beta_0$$

where:

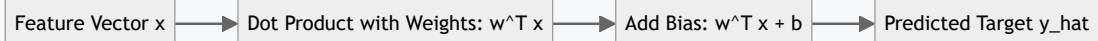
- w (or β_1) is the **weight / slope**, representing the change in $\hat{y}$ per unit change in x .
- b (or β_0) is the **bias / intercept**, representing the predicted value of y when $x = 0$.

2. Multiple Linear Regression (p Features):

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_p x_p + b = \mathbf{w}^T \mathbf{x} + b$$

In compact matrix notation for N observations:

$$\hat{\mathbf{y}} = X\mathbf{w} + b\mathbf{1}$$



2. Geometric Interpretation Across Dimensions

- **1 Feature ($p = 1$):** The model fits a **Straight Line** in 2D Euclidean space ($\mathbb{R}^2$).
- **2 Features ($p = 2$):** The model fits a **2D Plane** in 3D Euclidean space ($\mathbb{R}^3$).
- **p Features ($p \geq 3$):** The model fits a **(p)-dimensional Hyperplane** in $\mathbb{R}^{p+1}$.

3. Interpreting Model Parameters

Consider an insurance charges prediction model:

$$\text{charges} = 250 \cdot (\text{age}) + 330 \cdot (\text{bmi}) + 23800 \cdot (\text{smoker}) - 11000$$

1. **Slope for age (+250):** Holding all other features constant, each additional year of age increases predicted healthcare charges by \$250\$.
2. **Slope for smoker (+23800):** Smokers are predicted to incur \$23,800\$ more in annual medical charges than non-smokers.
3. **Intercept (-11000):** Baseline offset adjustment.

4. Python Implementation: Visualizing a 1D Linear Hypothesis

```
import numpy as np
import matplotlib.pyplot as plt

# Generate synthetic linear trend
np.random.seed(42)
x = np.linspace(10, 60, 50) # Age
y = 250 * x + 2000 + np.random.randn(50) * 1500 # Charges

# Specific parameters
w, b = 250.0, 2000.0

plt.figure(figsize=(8, 5))
plt.scatter(x, y, color='#1f77b4', alpha=0.7, label='Observed Data Points (x, y)')
plt.plot(x, w * x + b, color='#d62728', lw=2.5, label=f'Hypothesis Line: y = {w}x + {b}')

plt.title('Simple Linear Regression: Hypothesis Line', fontsize=12, fontweight='bold')
plt.xlabel('Age (Years)', fontsize=11)
plt.ylabel('Charges ($)', fontsize=11)
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend()
plt.show()
```